

AI ESSENTIALS: CLASS NOTES

Part I — Chapters 01 to 03

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Key Definitions

Artificial Intelligence (AI): Computer programs created to do smart tasks that normally require human thinking and decision-making.

Narrow AI (ANI): AI built to do one single job. It cannot work outside its design. Examples: social media feeds, Face Unlock, Google Maps traffic predictions, spam filters, spell-check.

General AI (AGI): A theoretical AI with human-level reasoning and learning across any topic. It does not exist today. It is only an idea for the future.

Generative AI (GenAI): AI trained on large amounts of data to create brand-new content (text, pictures, audio, or code) instead of just organizing old information.

HOW IT WORKS

Generative AI does not think or have feelings. It predicts the most likely next word or picture pattern based on what it learned.

Key Terms for This Course

Term	Meaning
Prompt	The message or instruction you type to an AI tool. Clearer prompts get better answers.
Hallucination	When AI gives a confident answer that is wrong or made up.
Free tier	The free plan of an AI tool. It usually has limits on messages, models, or features.
Placeholder	A stand-in like [CNIC] that you type instead of your real private information.
Source	Where information came from (a web page, a document). Search AI shows its sources.

Chapter 01: What Is Generative AI?

AI You Already Use Every Day

- **Social media feeds** learn what you watch and show similar videos.
- **Face Unlock** scans your face geometry to open your phone.
- **Maps** check live traffic to give the fastest route.
- **Text and email** block junk messages and fix typos.

How Generative AI Works

1. **Training:** it reads millions of books, articles, code, and pictures.
2. **Learning patterns:** it learns grammar, styles, and logic.
3. **Predicting:** it guesses the best next word or pixel, step by step.

What It Is NOT, and What You Should Do

GenAI is NOT...	So you should...
Conscious. It does not know or understand like a human.	Treat it as a tool that predicts, not a person who knows.
Always right. Confident answers can be wrong (hallucination).	Check important facts against a second source.
A mind-reader.	Write clear prompts: say what you want, for whom, and how long.

The AI Toolkit

Tool	Best used for	Website
ChatGPT	Quick answers, everyday writing, brainstorming.	chatgpt.com
Claude	Long writing tasks and careful reasoning.	claude.ai
Gemini	Google apps integration, web search, and photos.	gemini.google.com

The 3 Core AI Moves

- **Ask:** get information. "What is a spreadsheet used for?"
- **Explain:** simplify a tough topic. "Explain VLOOKUP like I am new to Excel."
- **Rewrite:** fix grammar or change tone. "Rewrite this email to sound more polite."

CHAPTER 01 RECAP

1. AI is already in your phone. GenAI creates new content by predicting the next word or pixel.
2. It is not conscious, not always right, and not a mind-reader.
3. Ask clearly, then check anything important.

Chapter 02: The AI Landscape

The 5 Types of AI Tools

Type	Best for	Example
Chat AI	Talking, writing, brainstorming	ChatGPT, Claude, Gemini
Search AI	Live web information with citations	Perplexity
Design AI	Images, posters, designs	Canva
Docs AI	Reading and summarizing uploaded files and PDFs	NotebookLM
Coding AI	Writing and fixing computer code	(code assistants)

Chat AI vs. Search AI

	Chat AI	Search AI
Answers from	What it learned in training. Some chat tools can also search the web when that feature is on.	Live web pages, searched first.
Sources	May not show where an answer came from.	Shows exactly where it found the information.
Best for	Writing, explaining, brainstorming.	Current news, prices, facts you must prove.

REMEMBER

Need something current, or something you must prove? Use a tool that shows its sources, and open the source to check it.

CHAPTER 02 RECAP

1. There are 5 types of AI tools: Chat, Search, Design, Docs, Coding.
2. Pick the type that matches the job.
3. Chat AI is for creating and explaining. Search AI is for finding and proving.

Chapter 03: Account Setup, Free Limits & Safety Basics

Try It in Class: No Login Needed

Lab computers cannot sign into personal accounts, so we practise on **Duck.ai** (duck.ai): free, anonymous AI chat with no account, email, or credit card. The six practice prompts are the three core moves in more detail:

Type	Example prompt	Core move
Explain	"Explain why the sky is blue, in two simple sentences."	Explain
Fix	"Fix the grammar: 'he go to school yesterday and study english'."	Rewrite
Summarize	"Summarize this in 2 lines: [short public paragraph]."	Rewrite
Write	"Write a two-line poem about rain."	Ask
Brainstorm	"Suggest three names for a new tea shop in Dadu."	Ask
Plan	"Give me a simple 3-step plan to revise for a test tomorrow."	Ask

The 4 Parts of an AI Screen

Part	What it does	Login?
New Chat button	Starts a fresh conversation with no memory of the last chat.	No
Message box	Where you type your prompt, at the bottom of the screen.	No
Chat history	Side menu that saves your past chats.	Yes
Account / Settings	Shows your profile and current plan.	Yes

Free Limits

Running AI costs companies real money, so free plans may limit: number of messages, advanced models, image generation, file uploads, and daily or monthly use. If you hit a limit, wait for the reset or try another tool. Spend free usage on useful tasks, not tests.

CHECK, DO NOT MEMORIZE

Limits change often. Always check the tool's own site or account page instead of trusting a number in your notes.

The Golden Safety Rule

GOLDEN SAFETY RULE

NEVER type or paste passwords, B-Form / CNIC numbers, home addresses, or personal information into free AI tools.

If you wouldn't write it on a public notice board at school, do not type it into an AI tool. This is true even when there is no login.

Why? Many free tools may store your chats, and some may use them to improve the service. Policies differ, and you cannot take a message back after you send it.

Keep private: passwords, CNIC / ID numbers, phone numbers when unnecessary, home addresses, private student information, private documents, and anyone else's confidential information.

Make It Safe with Placeholders

Unsafe	Safer
"Rewrite this letter. My CNIC is 42101-1234567-1."	"Rewrite this letter. My CNIC is [CNIC]."

Real information	Type this instead
Password	[PASSWORD]
CNIC / ID number	[CNIC]
Phone number	[PHONE]
Home address	[ADDRESS]
Student's name	[STUDENT NAME]
Private document details	[PRIVATE INFO]

The AI can still help with the structure of your task without seeing the real details. When in doubt, leave it out.

Homework: Set Up Your Own Account

Do this at home, on your own device with your own email:

1. Go to the tool's official website (chatgpt.com, claude.ai, or gemini.google.com).
2. Sign up with your email or a Google account.
3. Verify your email or phone number if asked.
4. Log in and start chatting.

Then: find Chat History and Account / Settings, and check your plan's free limit.

CHAPTER 03 RECAP

1. You can practise with no account. Setup at home is 4 steps.
2. Every AI screen has New Chat, Message Box, Chat History, and Account / Settings.
3. Free plans have limits. Check, do not guess.
4. Never share private information. Use [PLACEHOLDERS].

Test Yourself

1. What does the New Chat button do?

- A. Deletes your account
- B. Starts a fresh conversation
- C. Changes your password
- D. Shows your free limit

2. Which should NOT be pasted into an AI tool?

- A. A general question
- B. A public paragraph
- C. Your password
- D. A fictional story

3. Which is the safest way to ask?

- A. "My CNIC is 42101-1234567-1. Fix my form."
- B. "My CNIC is [CNIC]. Fix my form."
- C. "My CNIC is 4210112345671. Fix my form."

4. Which of these is safe to paste into an AI tool?

- A. Your home address
- B. A paragraph from a public news article
- C. A classmate's phone number
- D. Your student ID

5. A hallucination is when AI...

- A. Refuses to answer
- B. Gives a confident answer that is wrong or made up
- C. Runs out of free messages
- D. Deletes your chat

6. You need today's cricket score with a source link. Which tool type fits best?

- A. Design AI
- B. Docs AI
- C. Search AI
- D. Coding AI

Say it out loud:

- 7. Why can free AI plans have usage limits?
- 8. Name three parts of an AI screen. Which need a login?
- 9. Rewrite this safely: "Please improve this application. My CNIC is 42101-1234567-1."

Answer Key

Q	Answer
1	B. Starts a fresh conversation.
2	C. Your password.
3	B. Replace the real number with a placeholder. The AI still helps with the structure.
4	B. Anything already public is fine. A, C and D are private information.
5	B. A confident answer that is wrong or made up. Check important facts.
6	C. Search AI finds live information and shows sources.
7	Running AI needs computing power that costs the company real money, so free use is limited.
8	New Chat and Message Box (no login). Chat History and Account / Settings (login needed).
9	"Please improve this application. My CNIC is [CNIC]." Or leave the CNIC out completely.